

Complete Design Report

The full concept and preliminary design proposal.

CONTENTS

- Al Safa 2 Park Complete Design Report
- Board 2 evidence
- Fig10 masterplan
- Fig11 cost plan
- Section

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository	github.com/wasim437/AI_SAFA
Live portal	wasim437.github.io/AI_SAFA/
Produced by	run_analysis.py · src/plan.py · src/costing.py · models/headline_metrics.json
Reproduce	<code>python run_analysis.py</code> · <code>python -m tests.test_pipeline</code>

How this document was produced

This submission is the output of a ten-phase process in which each phase is checked against the one before it. Nothing downstream is hand-drawn or hand-typed: change an input, re-run, and every chart, drawing and figure moves with it — or a test fails loudly. The phases behind this particular document are marked.

Phase 1 — Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

Phase 2 — Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

Phase 3 — Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

Phase 4 — Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

Phase 5 — Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

Phase 6 — Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

Phase 7 — Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

Phase 8 — User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

Phase 9 — AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

Phase 10 — Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

THE CODE AND DATA BEHIND THIS DOCUMENT

[run_analysis.py](#)

[src/plan.py](#)

[src/costing.py](#)

[models/headline_metrics.json](#)

REPRODUCE IT

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 38 checks
```

Complete Design Report

Falaj Al Safa — concept and preliminary design for Al Safa 2 Park

A 15,000 m² neighbourhood park in Dubai, redesigned around one question: how do you make an outdoor space usable in a city that is too hot to stand in? The answer proposed here raises comfortable daylight hours from 44.5% to 64.6%, within a capital cost of AED 27.0 M against the AED 35 M budget.

Upload slot 10 — Complete Design Report

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai

Mohamed Wasim · Individual Applicant

Issued 03 August 2026

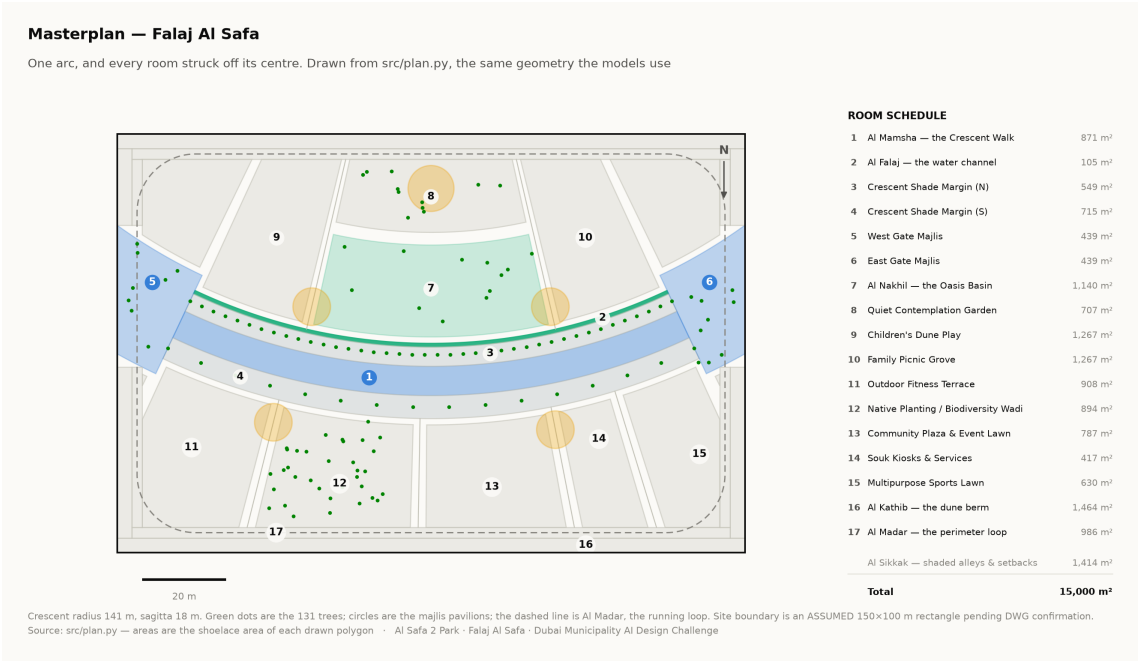
Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. The scheme

One arc. A crescent of shade 141 m in radius sweeps across the site, bowing convex south so that its hollow opens to the north. A 0.9 m water channel runs beneath its northern drip line. Every room in the park is struck off the same centre, so no room is a rectangle and every room faces the crescent square-on.

Element	What it is	Description
Al Hilal	the Crescent Canopy	18 m gridshell at 4.5 m over a 7 m walk, with a 3 m southern louvre
Al Falaj	the water channel	0.9 m wide, on the canopy's drip line so it is shaded all day and evaporates less
Al Nakhil	the Oasis Basin	a sunken palm court held in the crescent's concave side
Al Sikkak	the alleys	radial — each one a radius of the same arc
Al Madar	the perimeter loop	a tree-shaded running and walking circuit
Al Kathib	the dune berm	planted earth against the roads — noise, glare and heat

The six named elements. All are generated from a single arc definition in src/plan.py.



Masterplan. Technical drawing — to scale.

2. Performance

Measure	Today	As designed
Comfortable daylight hours	44.5%	64.6%
Peak heat index	56.8 °C	48.7 °C
Crescent Walk shaded	—	87.3%
Site-wide mean shade	—	34.1%
Trees	—	131

3. Room schedule

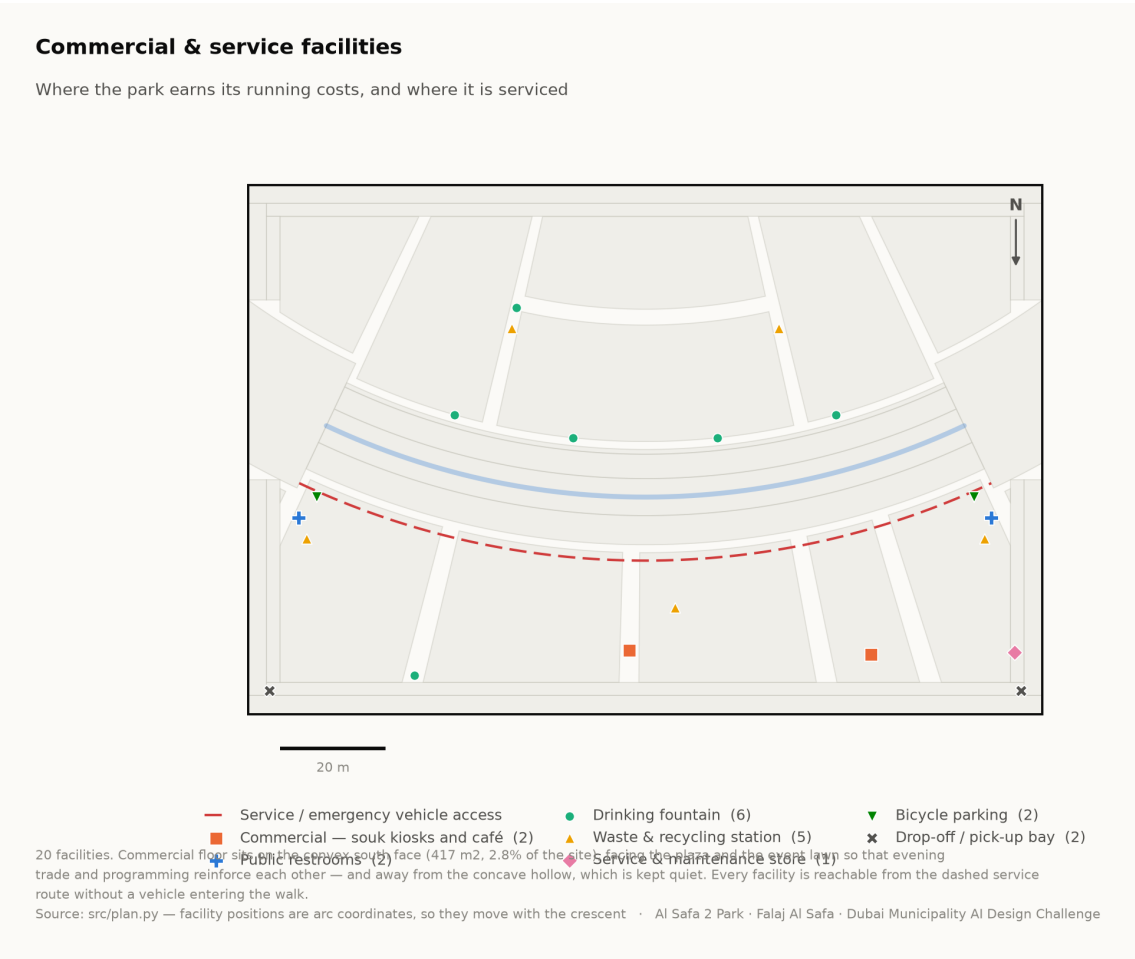
Room	Category	Area (m²)	% of site
Al Mamsha — the Crescent Walk	Circulation	871	5.81%
Al Falaj — the water channel	Water	105	0.70%
Crescent Shade Margin (N)	Green	549	3.66%
Crescent Shade Margin (S)	Green	715	4.77%
West Gate Majlis	Arrival	439	2.93%
East Gate Majlis	Arrival	439	2.93%
Al Nakhil — the Oasis Basin	Green	1,140	7.60%
Quiet Contemplation Garden	Passive	708	4.72%
Children's Dune Play	Active	1,267	8.45%
Family Picnic Grove	Passive	1,267	8.45%
Outdoor Fitness Terrace	Active	908	6.06%
Native Planting / Biodiversity Wadi	Green	894	5.96%
Community Plaza & Event Lawn	Social	788	5.25%
Souk Kiosks & Services	Commercial	417	2.78%
Multipurpose Sports Lawn	Active	630	4.20%
Al Kathib — the dune berm	Green_Buffer	1,464	9.76%
Al Madar — the perimeter loop	Circulation	986	6.57%
Al Sikkak — shaded alleys & setbacks	Circulation	1,414	9.43%

Total 15,000 m².

4. Circulation, accessibility and facilities



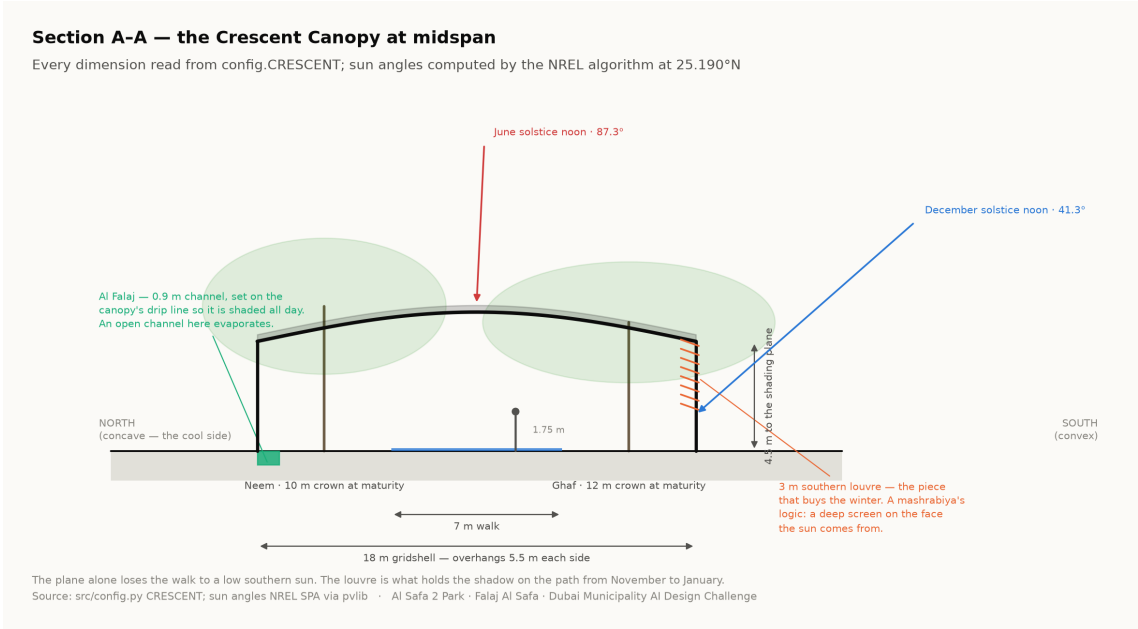
Circulation and accessibility. Technical drawing — to scale.



Commercial and service facilities, with the service route. Technical drawing — to scale.

20 facilities are placed against the drawn geometry: 2 commercial, 2 restroom blocks, 6 drinking fountains, 5 waste and recycling points, 1 maintenance store, 2 bicycle parking areas and 2 drop-off bays. Each is reachable from the service route without a vehicle entering the walk.

5. The section that makes it work



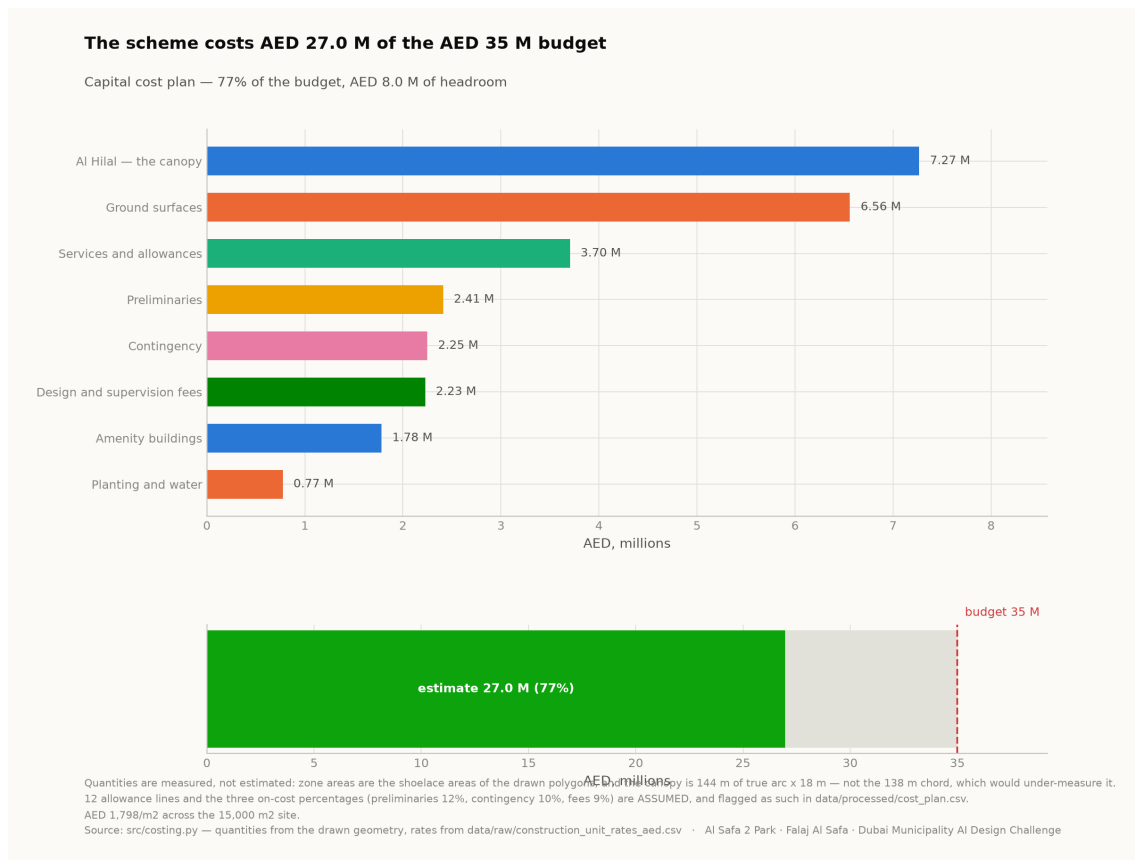
Section A–A: a 7 m walk under an 18 m gridshell at 4.5 m, with a 3 m southern louvre. Sun angles computed, not drawn. Technical drawing — to scale.

6. Feasibility — cost against the AED 35 M budget

The capital cost plan measures quantities from the drawn geometry rather than estimating them. Zone areas are shoelace areas; the canopy is priced on 144.2 m of *true arc* — not the 138 m chord, which would under-measure the structure by more than 6 m.

Element	AED millions
Al Hilal — the canopy	7.27
Ground surfaces	6.56
Services and allowances	3.70
Amenity buildings	1.78
Planting and water	0.77
Preliminaries	2.41
Contingency	2.25
Design and supervision fees	2.23
TOTAL	26.97
Budget	35.00
Headroom	8.03

Total AED 26,973,013, or 77.1% of the budget, at AED 1,798/m² across the site. Allowance lines and the three on-cost percentages are flagged ASSUMED in data/processed/cost_plan.csv.



Capital cost plan against the budget. Analysis output — computed from project data.

7. Honesty notes

- The **site boundary is assumed** — a 150 × 100 m rectangle pending confirmation against the issued CAD file. Every area figure depends on it.
- The **climate series is modelled** from 39 years of monthly normals, verified back to within 0.39 °C, and labelled as modelled everywhere it appears.
- **Visitor demand is a scenario**, not a prediction, and is deliberately excluded from the machine learning suite.
- **Diffuse radiation is excluded** from the shade model, so shaded comfort is stated conservatively.
- The **photovoltaic array runs a deficit**, covering about 13% of load — not a surplus.
- The **renders are illustrations**, not photographs of a built thing, and none is presented as analysis.

8. Corrections this submission makes to itself

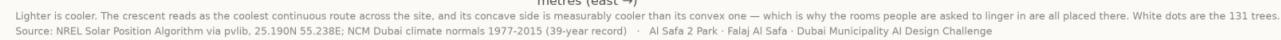
An earlier version of this project claimed 99.2% annual shade. It did not survive a geometric check and was withdrawn; the re-solved section measures 87.3%. Three visuals that presented invented data as measurement were also withdrawn. Both corrections are stated here rather than quietly removed, because a submission that audits itself in public is more credible than one that never had to.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Presentation board.

Dubai Municipality AI Park Design Challenge | Board 2 of 2 — EVIDENCE

July afternoon heat index per square metre — the shade strategy, tested



Permutation importance — the drop in accuracy when each feature is shuffled



Drawn at mature canopy radius. The avenue is structural; it carries the shoulder seasons the gridshell cannot



28 of 131 trees are UAE natives (Ghaf and Date Palm). Peak summer irrigation demand 11.8 m³/day. Ghaf takes the southern rank because it is the most drought-tolerant species in the schedule and that rank is the more exposed of the two. Source: data/eachspecies_water_carbon_rates.csv; layout from src/scenery - AI Safe 2 Park - Jalal Al Safa - Dubai Municipality AI Design Challenge

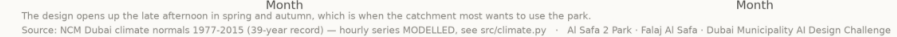
Share of the year's 4,402 daylight hours in each comfort band



Ray-traced ground-plane occlusion, 1 m grid, sampled across the year



Mean heat index by hour and month — exposed today, shaded as designed



Elevation — the Crescent Canopy
60 m of the 124 m run, to scale, 6 m structural bay



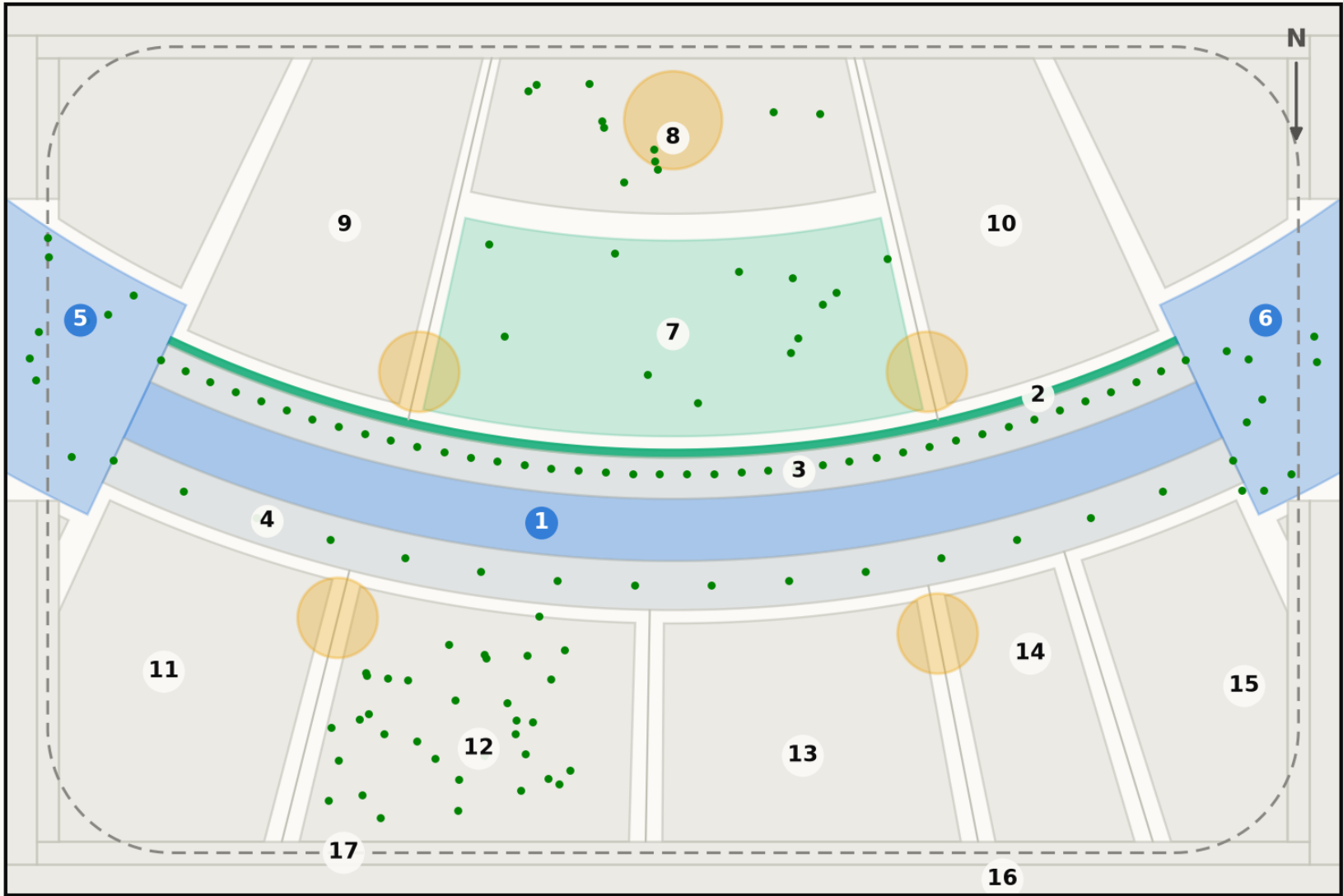
Shade surrogate R^2 0.994 · comfort classifier 97.5% with temperature and humidity withheld · 2 microclimate regimes, k selected by silhouette

Fig10 masterplan

Technical drawing — to scale. Geometry generated by src/plan.py; every area is the measured area of the drawn polygon.

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
	Al Sikkak — shaded alleys & setbacks	1,414 m²

Total 15,000 m²

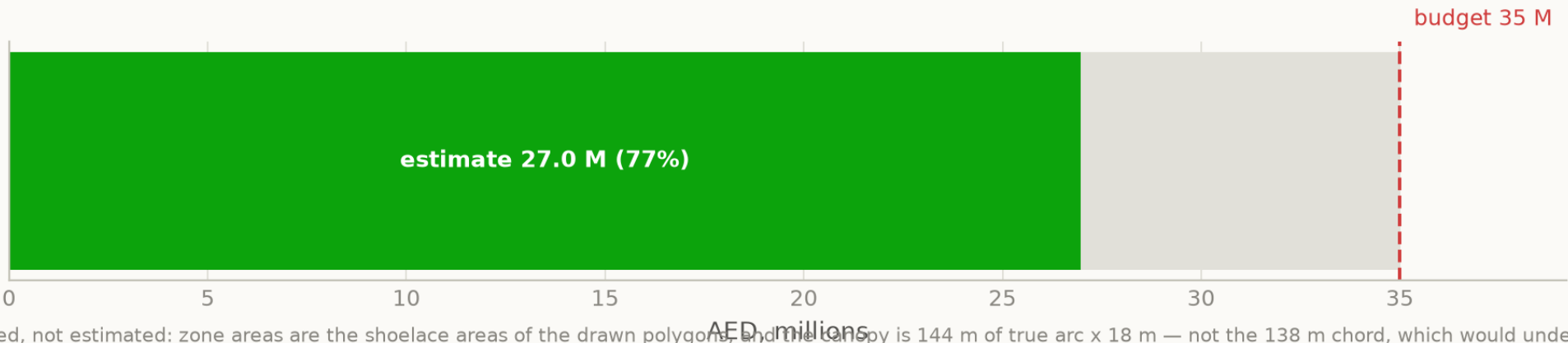
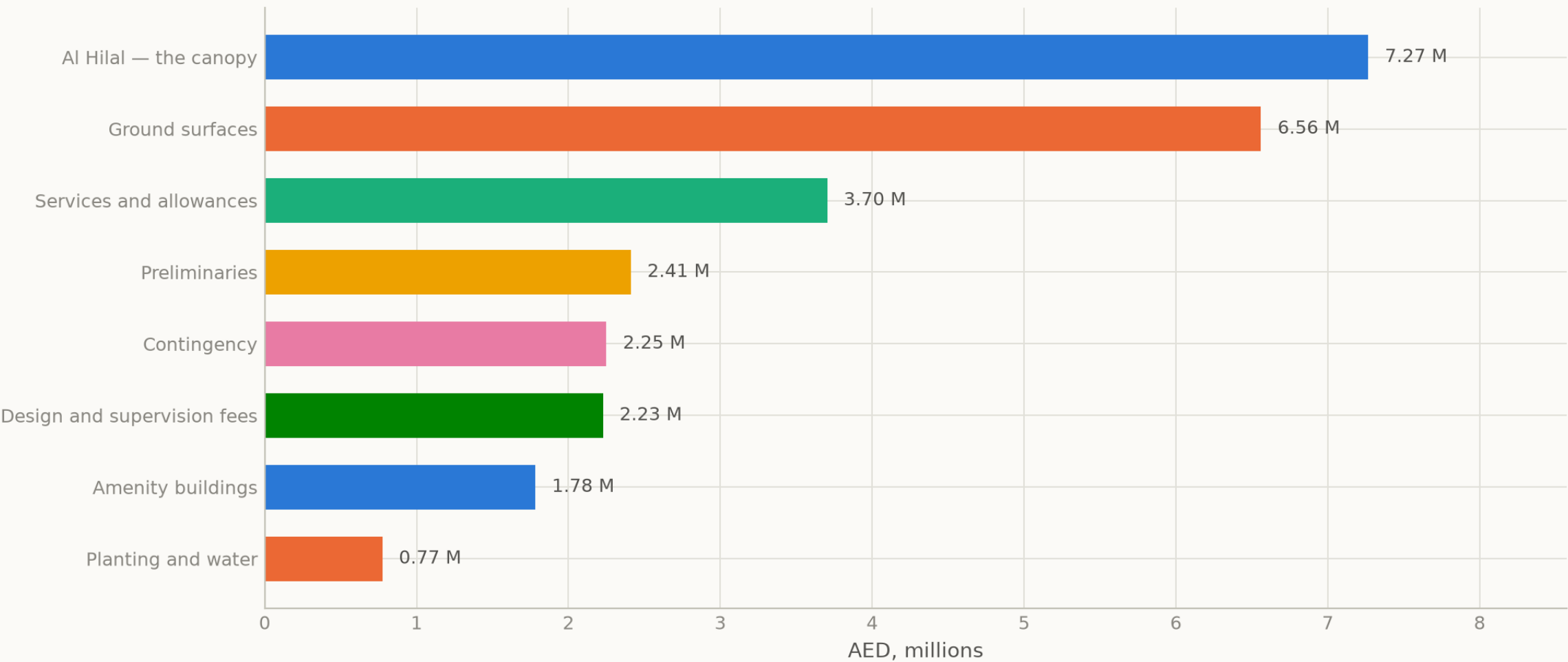
Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig11 cost plan

Analysis output — computed from project data.

The scheme costs AED 27.0 M of the AED 35 M budget

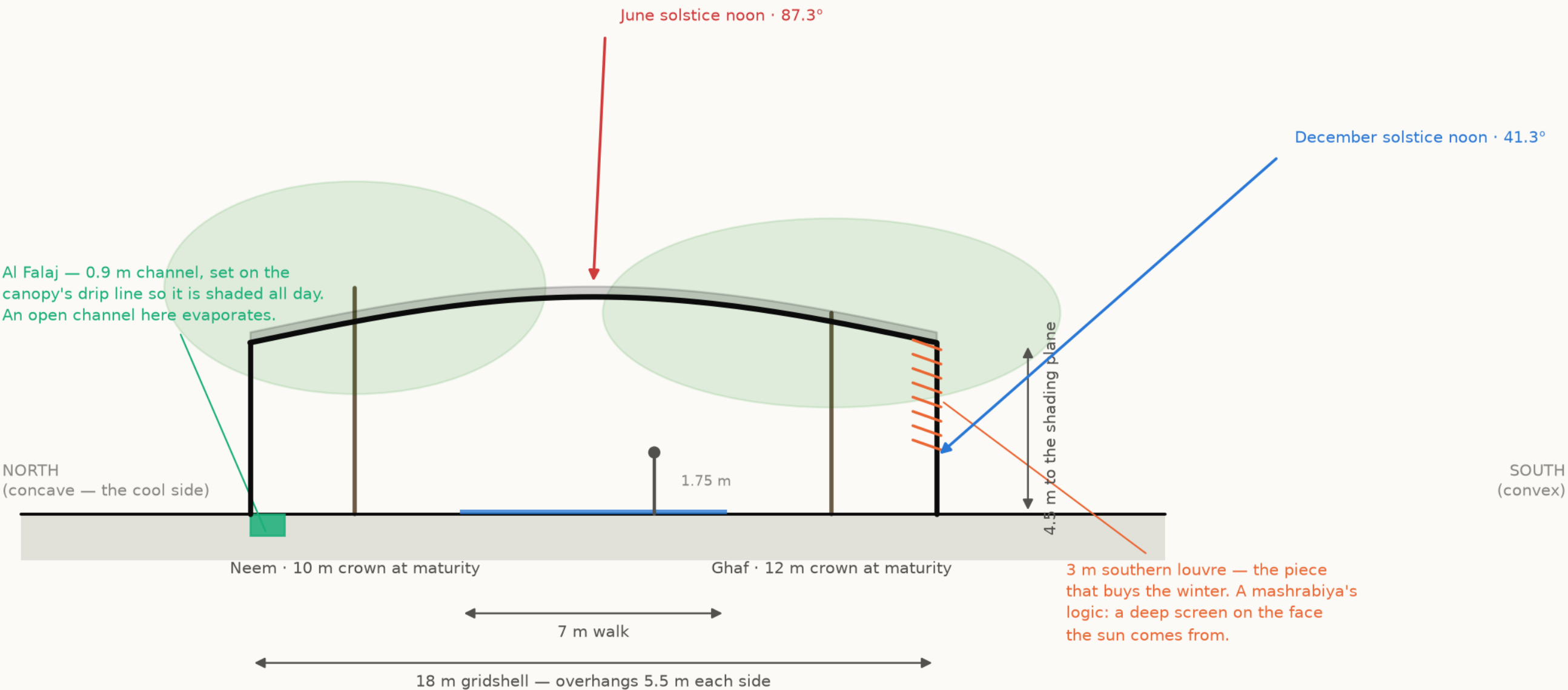
Capital cost plan — 77% of the budget, AED 8.0 M of headroom



Quantities are measured, not estimated: zone areas are the shoelace areas of the drawn polygons, and the canopy is 144 m of true arc x 18 m — not the 138 m chord, which would under-measure it. 12 allowance lines and the three on-cost percentages (preliminaries 12%, contingency 10%, fees 9%) are ASSUMED, and flagged as such in data/processed/cost_plan.csv. AED 1,798/m2 across the 15,000 m2 site. Source: src/costing.py — quantities from the drawn geometry, rates from data/raw/construction_unit_rates_aed.csv · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Section A-A — the Crescent Canopy at midspan

Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N



The plane alone loses the walk to a low southern sun. The louvre is what holds the shadow on the path from November to January.
Source: src/config.py CRESCENT; sun angles NREL SPA via pvlib · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The complete project, and where to check it

Every one of the 129 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Masterplan
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Diagrams
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (8)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
PUSH_TO_GITHUB.md	Project document
README.md	The design argument, written for a juror
RENDER_PROMPTS.md	Project document
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data
figures/fig05_surrogate_performance.png	Analysis output — computed from project data
figures/fig06_feature_importance.png	Analysis output — computed from project data
figures/fig07_confusion_matrix.png	Analysis output — computed from project data
figures/fig08_microclimate_regimes.png	Analysis output — computed from project data
figures/fig09_diurnal_comfort.png	Analysis output — computed from project data
figures/fig10_masterplan.png	Analysis output — computed from project data
figures/fig11_cost_plan.png	Analysis output — computed from project data

The complete project — continued

THE TECHNICAL DRAWINGS AND BOARDS (7)

[design/visuals/circulation_crescent.png](#)
[design/visuals/elevation_crescent.png](#)
[design/visuals/facilities_crescent.png](#)
[design/visuals/planting_crescent.png](#)
[design/visuals/section_crescent.png](#)
[design/boards/board_1_concept.png](#)
[design/boards/board_2_evidence.png](#)

Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)

THE ANALYSIS CODE (13)

[src/__init__.py](#)
[src/boards.py](#)
[src/climate.py](#)
[src/config.py](#)
[src/costing.py](#)
[src/dataset.py](#)
[src/drawings.py](#)
[src/figures.py](#)
[src/models.py](#)
[src/plan.py](#)
[src/solar.py](#)
[src/viz.py](#)
[run_analysis.py](#)

Analysis module
The two presentation boards
The 8,760-hour year rebuilt from NCM normals
Every constant the project reads
The cost model against the AED 35 M ceiling
Assembles the ML training tables
Section, elevation, circulation, planting, facilities
The analysis charts
The four models
SINGLE SOURCE of the crescent geometry
Sun position and shadow ray-tracing
Analysis module
Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (11)

[tools/__init__.py](#)
[tools/build_docs.py](#)
[tools/build_links.py](#)
[tools/build_notebook.py](#)
[tools/build_reports.py](#)
[tools/build_submission_pdfs.py](#)
[tools/organize_repo.py](#)
[tools/report_content.py](#)
[tools/sync_film.py](#)
[tools/sync_portal.py](#)
[tools/sync_submission.py](#)

Build tool
Build tool
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Build tool
Build tool
Build tool

THE DATA, AS ISSUED (7)

[data/raw/construction_unit_rates_aed.csv](#)
[data/raw/dubai_climate_normals_ncm.csv](#)
[data/raw/dubai_population_dsc_2023.csv](#)
[data/raw/site_zoning_schedule.csv](#)
[data/raw/sources.json](#)
[data/raw/species_water_carbon_rates.csv](#)
[data/raw/walk_catchment_rings.csv](#)

Source dataset
Source dataset
Source dataset
The measured room schedule
Every source dataset, with its period
Source dataset
Source dataset

THE DATA, AS PROCESSED (6)

[data/processed/cost_plan.csv](#)
[data/processed/hourly_climate_comfort_8760.csv](#)
[data/processed/masterplan_geometry.json](#)
[data/processed/planting_layout.csv](#)
[data/processed/schema.json](#)
[data/processed/spatial_grid_comfort.csv](#)

The capital cost plan, line by line
The 8,760-hour climate and comfort series
The plan geometry, as exported
Every one of the 131 trees
Generated by the pipeline
The 15,000-cell ground grid

The complete project — continued

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	38 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFI_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded
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THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Ai Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD PARKS MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/09_Material_Landscape_Palette/MANIFEST.md	What is in the slot, and what produced each file
submission/10_Complete_Design_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md	What is in the slot, and what produced each file
submission/12_Concept_Animation_Video/MANIFEST.md	What is in the slot, and what produced each file

WORKING HISTORY (9)

archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why
archive/phases/02_PHASE2_PROBLEM_DEFINITION/README.md	Working record
archive/phases/03_PHASE3_OPPORTUNITY_AND_OBJECTIVES/README.md	Working record
archive/phases/04_PHASE4_CONCEPT_DEVELOPMENT/README.md	Working record
archive/phases/05_PHASE5_MASTERPLAN_DEVELOPMENT/README.md	Working record
archive/phases/06_PHASE6_DETAILED_DESIGN/README.md	Working record
archive/phases/07_PHASE7_PERFORMANCE_AND_SUSTAINABILITY/README.md	Working record
archive/phases/08_PHASE8_USER_EXPERIENCE_AND_ACTIVATION/README.md	Working record
archive/phases/09_PHASE9_AI_WORKFLOW_AND_VISUALIZATION/README.md	Working record

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.